

if_elif_else

```
In [1]: if True: # indentation is always 4 spaces
        print('Data Science')
```

Data Science

```
In [3]: if True:
        print('ds')
```

ds

```
In [4]: if False:
        print('Data Science')
        print('bye for now')
```

bye for now

```
In [5]: if True: # indentation is always 4 spaces
        print('Data Science')
        print('bye for now')
```

Data Science

bye for now

```
In [6]: if False: # indentation is always 4 spaces
        print('Data Science')
        else:
        print('bye for now')
```

bye for now

```
In [7]: #to print only even number
x = 4
r = x % 2

if r == 0:
    print('Even number')
```

Even number

```
In [8]: #to print only even number
x = 5
r = x % 2

if r == 0:
    print('Even number')
```

```
In [9]: x = 5
r = x % 2

if r == 0:
    print('Even number')
print('odd number')
```

odd number

```
In [10]: x = 6
r = x % 2
```

```
if r == 0:  
    print('Even number')  
print('odd number')
```

Even number
odd number

```
In [11]: x = 10  
        r = x % 2  
  
        if r == 0:  
            print('Even number')  
        if r == 1:  
            print('odd number')
```

Even number

```
In [12]: x = 11  
        r = x % 2  
  
        if r == 0:  
            print('Even number')  
        if r == 1:  
            print('odd number')
```

odd number

```
In [13]: x = 11  
        r = x % 2  
  
        if r == 0:  
            print('Even number')  
        else:  
            print('odd number')
```

odd number

```
In [14]: x = 23  
        r = x % 2  
  
        if r == 0:  
            print('Even number')  
        if r == 1:  
            print('odd number')
```

odd number

```
In [15]: x = 17  
        r = x % 2  
  
        if r == 0:  
            print('Even number')  
  
        if r != 0:  
            print('odd number')
```

odd number

```
In [16]: x = 5  
        r = x % 2
```

```
if r == 0:  
    print('Even number')  
else:  
    print('Odd Number')
```

Odd Number

```
In [17]: x = 4  
         r = x % 2  
  
         if r == 0:  
             print('Even number')  
         else:  
             print('Odd Number')
```

Even number

```
In [18]: x = 3  
         r = x % 2  
  
         if r == 0:  
             print('Even number')  
             if x>5:  
                 print('greater number')  
         else:  
             print('Odd Number')
```

Odd Number

```
In [19]: x = 4  
         r = x % 2  
  
         if r == 0:  
             print('Even number')  
             if x>5:  
                 print('greater number')  
         else:  
             print('Odd Number')
```

Even number

```
In [20]: x = 4  
         r = x % 2  
  
         if r == 0:  
             print('Even number')  
             if x>5 :  
                 print('greater number')  
             else:  
                 print('not greater ')  
         else:  
             print('odd number')
```

Even number

not greater

```
In [21]: x = 6  
         r = x % 2  
  
         if r == 0:
```

```
print('Even number')
if x>5 :
    print('greater number')
else:
    print('not greater ')

else:
    print('odd number')
```

Even number
greater number

In [22]: *# when you use if it will check all condition but if we mention elif then it won't
when we use if condition it will check all every block of code better debug i
you can debug with value 1 & 2 for both if & elif*

```
x = 2

if x == 1:
    print('one')
if x == 2:
    print('Two')
if x == 3:
    print('Three')
if x == 4:
    print('four')
```

Two

In [23]: *# elif it won't check till the block once you find the output it won't go to next
you can try with multiple parameter 1, 2 & 3 value in x*

```
x = 4

if(x == 1):
    print('one')
elif(x == 2):
    print('Two')
elif(x == 3):
    print('Three')
elif(x == 4):
    print('four')
```

four

In [24]: x = 7

```
if(x == 1):
    print('one')
elif(x == 2):
    print('Two')
elif(x == 3):
    print('Three')
elif(x == 4):
    print('four')
```

In [25]: x = 7

```
if(x == 1):
    print('one')
```

```
elif(x == 2):  
    print('Two')  
elif(x == 3):  
    print('Three')  
elif(x == 4):  
    print('four')  
  
else:  
    print('wrong output')
```

wrong output

```
In [26]: x = 4  
  
if(x == 1):  
    print('one')  
elif(x == 2):  
    print('Two')  
elif(x == 3):  
    print('Three')  
elif(x == 4):  
    print('four')  
else:  
    print('wrong output')
```

four

```
In [27]: x = 10  
  
if(x == 1):  
    print('one')  
elif(x == 2):  
    print('Two')  
elif(x == 3):  
    print('Three')  
elif(x == 4):  
    print('four')  
  
else:  
    print('wrong output')
```

wrong output

```
In [28]: #short hand if  
a = 30  
b = 20  
if a > b:  
    print("a is greater than b")
```

a is greater than b

```
In [ ]:
```